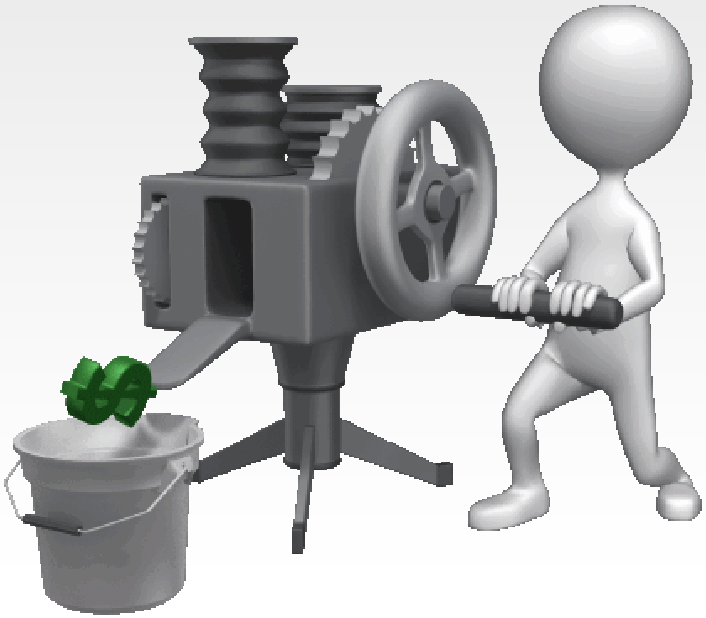


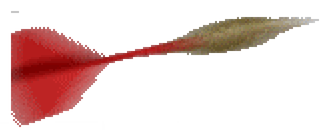
LECTURE 3

MANUFACTURING OPERATIONS

Contents

Objectives

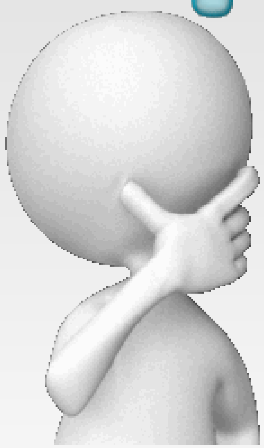
1. Outline the operation types in manufacturing systems
2. Define product-production relationship
3. Define the complexity of manufacturing processes
4. Formulate the production characteristics
5. Describe the cost and selling price relationship
6. Describe manufacturing strategies

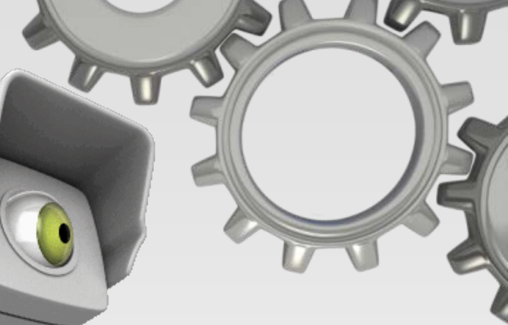


Review

1. Components of a manufacturing system
2. Types of manufacturing systems
3. Classification of manufacturing systems
4. Manufacturing progress function

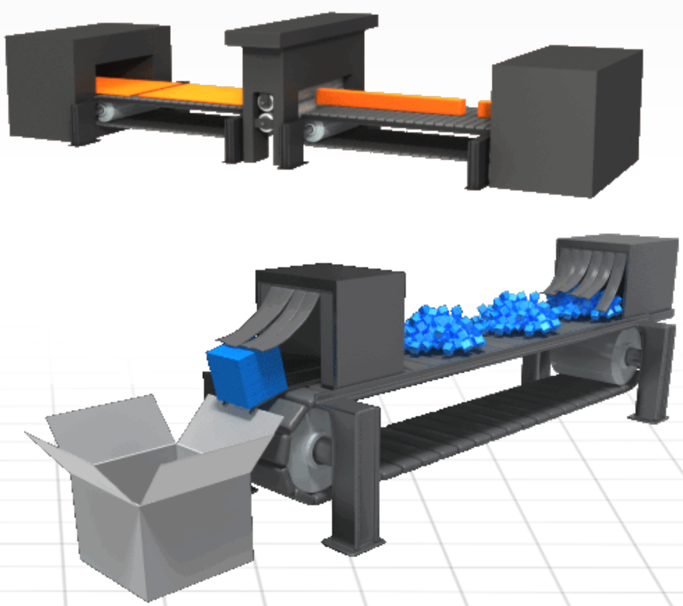
Operations

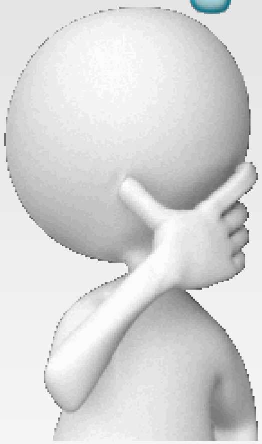




Types of operations:

- processing and assembly
- material handling
- inspection and test
- coordination and control



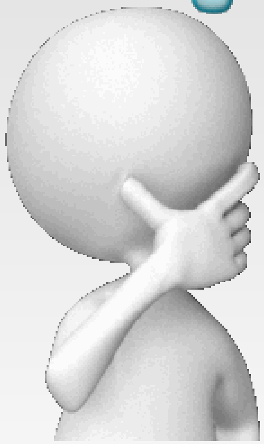


Processing operations

A processing operation transforms a work material from one state of completion to a more advanced state that is closer to the final desired part or product.

Processing operations may be grouped into:

- Shaping operations,
- Property enhancing operations,
- Surface processing operations.



Assembly operations

An assembly operation joins two or more components to create a new entity, which is called an assembly, subassembly, or some other term that refers to the specific joining process.

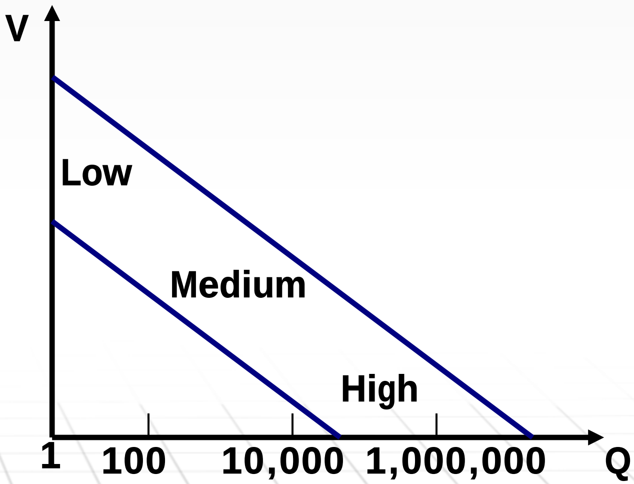
Components of the new entity can be connected together:

- **permanently** (welding, soldering, adhesive bonding, press fitting, expansion fits, and rivets)
- **semi-permanently** (threaded fasteners like screws, bolts, nuts)

Product-production relationship

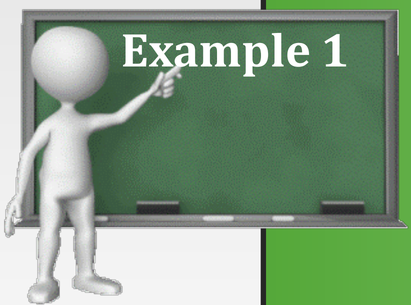
Companies organize their manufacturing operations and production systems accordingly with the products parameters they make:

- ➢ product quantity
- ➢ product variety:
  - ✓ hard
  - ✓ soft
- ➢ complexity of assembled products
- ➢ complexity of individual parts



▪ Figure 3.1 Relationship between product quantity and variety in discrete product manufacturing.

Production quantity and product variety



A company specializes in consumer photographic products. It produces only cameras and projectors. In its camera line it offers 3 models with hard variety, and in its projector line it offers 2 models with soft variety.

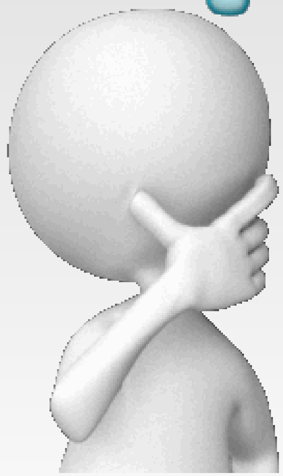
a) What is the totality of product models offered?
b) If the company produce 1000 of each camera model and 100 of each projector model what is the total quantity of all products made by company?

Solution:

a) $$V = \sum_{i=1}^{n} V_i = V_1 + V_2 = 3 + 2 = 5$$

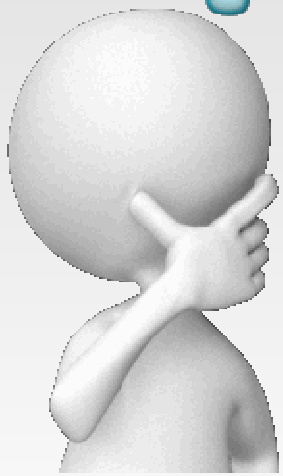
b) $$Q_t = \sum_{i=1}^{n} Q_i = Q_1 + Q_2 = c \times V_1 + p \times V_2 = 1000 \times 3 + 100 \times 2 = 3200$$

Product and part complexity



- For an assembled product, one possible indicator of its complexity is the number of components or parts (n_p).
- For a produced component, a possible measure of its complexity is the number of processing steps or operations (n_o) required to produce it.

Type of plant	**n_p and n_o values**	**Description**
Part producer	$n_p=1, n_0>1$	Individual components are produced. Each component requires multiple processing steps.
Assembly plant	$n_p>1, n_0=1$	A pure assembly plant produce no part. It is assumed that one operation is required to assemble each part to the product.
Vertically integrated plant	$n_p>1, n_0>1$	This plant makes all its parts and assemble them.



Product and part complexity

$$n_{pt} = \sum_{i=1}^{V} Q_i \times n_{pi}$$

$$n_{ot} = \sum_{i=1}^{V} Q_i \times n_{pi} \times \sum_{k=1}^{n_{pi}} n_{oik}$$

where:

n_{pt}-total number of parts made in the factory (pc/yr)

n_{pi}-number of parts in product i (pc/product)

Q_i-annual quantity of product style i (products/yr)

n_{ot}- total number of operations

n_{oik}- number of processing operations for each part *k*, for a product *i*

Self-assessment test

1. The variety and quantity of goods are in a relationship of direct proportionality. **T** **F**

2. A vertically integrated plant only assembles the products. **T** **F**

3. The complexity of a product is given by the number of components or the number of operations needed for manufacturing. **T** **F**

4. A dedicated manufacturer of assemblies produces also the components necessary for their products. **T** **F**

Production concepts

- ❑ Production rate
- ❑ Production capacity
- ❑ Utilization and availability (reliability)
- ❑ Manufacturing lead time
- ❑ Work-in-process

Production rate

The production rate for an individual processing or assembly operation is usually expressed as an hourly rate, parts or products per hour.

The operation cycle time T_c is defined as the time that one work unit spends being processed or assembled.

$$T_c = T_o + T_h + T_{th}$$

where:

T_o - actual machining operation time
T_h - workpart handling time
T_{th} - tool handling time per workpiece

The production rate depends on the type of production.

Production rate for batch production

$$T_b = T_{su} + Q \times T_c$$

$$T_p = \frac{T_b}{Q}$$

$$R_p = \frac{60}{T_p}$$

where:

- T_b – batch processing time (min)
- T_{su} – setup time to prepare for the batch (min)
- Q – batch quantity (pc)
- T_c – operation cycle time per work unit (min/cycle or min/pc)
- T_p – average production time per work unit
- R_p – hourly production rate (pc/hr)

The production rate depends on the production type.

Production rate for job shop production

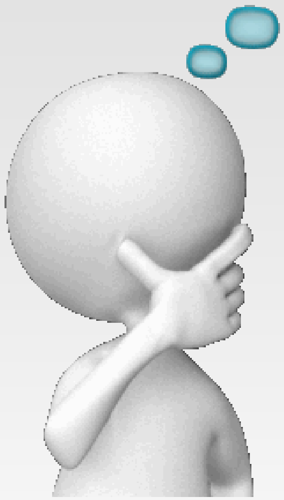
$$T_p = T_{su} + T_c \quad \text{if} \quad (Q = 1)$$

$$R_p = \frac{60}{T_p}$$

where:

- T_{su} – setup time to prepare for the batch (min)
- Q – batch quantity (pc)
- T_c – operation cycle time per work unit (min/cycle or min/pc)
- T_p – average production time per work unit
- R_p – hourly production rate (pc/hr)

The production rate depends on the production type.



Production rate for quantity type mass production

$$R_p \rightarrow R_c = \frac{60}{T_c}$$

where:

- T_c – operation cycle time (min/pc)
- R_p – hourly production rate (pc/hr)
- R_c – operation cycle rate of the machine (pc/hr)

The production rate depends on the production type.

Production rate for flow line mass production

$$T_c = T_r + MaxT_o$$

$$R_c = \frac{60}{T_c}$$

where:

- T_c – ideal cycle time (min/cycle)
- T_r – time to transfer work units between stations each cycle (min/pc)
- $MaxT_o$ – operation time at the bottleneck station (the maximum of the operation times for all stations on the line, min/cycle)
- R_c – theoretical or ideal production rate (cycle rate, cycles/hr)

The production rate depends on the production type.

Production capacity (1)

Production capacity is defined as the maximum rate of output that a production facility is able to produce under a given set of assumed operating conditions.

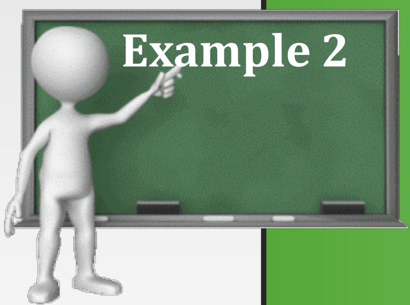
$$PC = n \times S \times H \times R_p$$

where:

- PC – production capacity of the facility (output units/wk)
- n – number of work centers producing in the facility
- S – number of shifts per period (shift/wk)
- H – hours per shift (hr)
- R_p – hourly production rate of each work center (output units/hr)

It is assumed that the units processed through the group of work centers are homogeneous, and therefore the value of R_p is the same for all units produced.

Production capacity



The turret lathe section has six machines, all devoted to the production of the same part. The section operates 10 shift/wk. The number of hours per shift averages 8.0. Average production rate of each machine is 17 unit/hr. Determine the weekly production capacity of the turret lathe section.

Solution:

$$\mathbf{PC = n \times S \times H \times R_p}$$

PC=6 x 10 x 8.0 x 17 = 8160 output unit/wk

Production capacity (2)

If it is included the possibility that each work unit is routed through n_0 operations, with each operation requiring a new setup on either the same or different machine, then:

$$PC = \frac{n \times S \times H \times R_p}{n_o}$$

What changes can be made to increase or decrease production capacity (plant capacity):

- over the short term:
  - change the number of shifts per week;
  - change the number of hours worked per shift.
- over the intermediate or longer term:
  - increase the number of work centers in the shop;
  - increase the production rate, by making improvements in methods or process technology;
  - reduce the number of operations required per work unit using combined operations, simultaneous operations, or integrations of operations.

Self-assessment test

1. The operation cycle time is the actual processing time. **T** **F**

2. The setup time is considered when determining the operation cycle time for job shop production. **T** **F**

3. Production capacity increases when the production rate increases. **T** **F**

4. For quantity type mass production, in the calculation of the operation cycle time, the system setup time must be included. **T** **F**

5. For the flow line mass production, the maximum operation time of all stations on the line is included in the calculation of the cycle time. **T** **F**

Utilization

Utilization refers to the amount of output of a production facility relative to its capacity.

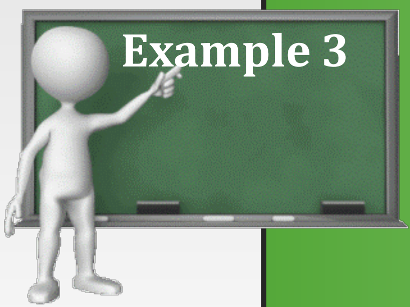
$$U = \frac{Q}{PC}$$

where:

U – utilization of the facility
Q – actual quantity produced by the facility during a given period
PC – production capacity for the same period

For convenience, utilization is often defined as the proportion of time that the facility is operating relative to the time available under the definition of capacity.

Utilization



A production machine operates 80 hr/wk (two shifts, 5 days) at full capacity. Its production rate is 20 unit/hr. During a certain week, the machine produced 1000 parts and was idle the remaining time. (a) Determine the production capacity of the machine. (b) What was the utilization of the machine during the week under consideration?

Solution:

$$\mathbf{PC} = \mathbf{n} \times \mathbf{S} \times \mathbf{H} \times \mathbf{R_p}$$

(a) PC=80 x 20=1600 unit/wk

$$\mathbf{U} = \frac{\mathbf{Q}}{\mathbf{PC}}$$

(b) U=1000/1600=0.625 or 62.5%

or:

H=(1000 pc)/(20 pc/hr)=50 hr (the time the machine was operated)

U=50/80=0.625

Availability

Availability is a common measure of reliability for equipment.

$$A = \frac{\textbf{MTBF-MTTR}}{\textbf{MTBF}}$$

where:

A	- availability
MTBF	- mean time between failures
MTTR	- mean time to repair

Availability is typically expressed as a percentage

Availability

Example 4

The turret lathe section has six machines, all devoted to the production of the same part. The section operates 10 shift/wk. The number of hours per shift averages 8.0. Average production rate of each machine is 17 unit/hr. The availability of the machines A=90%, and the utilization of the machines U=80%. Compute the expected plant output.

Solution:

$$\mathbf{Q = A \times U \times (n \times S \times H \times R_p)}$$

Q=0.90 x 0.80 x 6 x 10 x 8.0 x 17=5875 output unit/week

Manufacturing lead time (1)

Manufacturing lead time (MLT) is the total time required to process a given part or product through the plant.

$$MLT_j = \sum_{i=1}^{n_{oj}} \left(T_{suji} + Q_j \times T_{cji} + T_{noji} \right)$$

where:

MLT_j	– manufacturing lead time for part or product j (min)
T_{suji}	– setup time for operation i (min)
Q_j	– quantity of part or product j in the batch being processed
T_{cji}	– operation cycle time for operation i (min/pc)
T_{noji}	– nonoperation time associated with operation i (min)
i	– indicates the operation sequence in the processing; $i=1..n_{oj}$

Manufacturing lead time (2)

To simplify and generalize our model, let us:

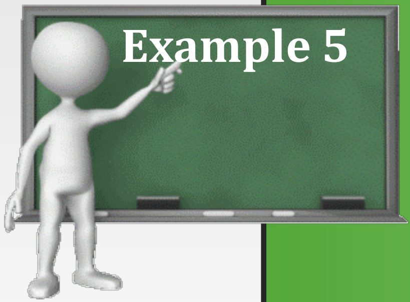
- assume that all setup times, operation cycle times and nonoperation times are equal for the n_{oj} machines;
- suppose that the batch quantities of all parts or products processed through the plant are equal and they are all processed through the same number of machines, so that $n_{oj}=n_o$

$$MLT = n_o \times \left(T_{su} + Q \times T_c + T_{no} \right)$$

where:

MLT – average manufacturing lead time for a part or product (min)
T_{su} – setup time (min)
Q_j – quantity of part or product j in the batch being processed
T_c – operation cycle time (min/pc)
T_{no} – nonoperation time (min)

Manufacturing lead time



A certain part is produced in a batch size of 100 units. The batch must be routed through five operations to complete the processing of the parts. Average setup time is 3 hr/operation, and average operation time is 6 min (0.1 hr). Average nonoperation time due to handling, delays, inspection, etc,. is 7 hours for each operation. Determine how many days it will take to complete the batch, assuming the plant runs one 8-hr shift/day.

Solution:

$$\mathbf{MLT} = \mathbf{n_o} \times \left(\mathbf{T_{su}} + \mathbf{Q} \times \mathbf{T_c} + \mathbf{T_{no}} \right)$$

MLT=5 x (3+100 x 0.1+7)=100 hr

At 8 hr/day, this amounts to 100/8=12.5 days

Work-in-process

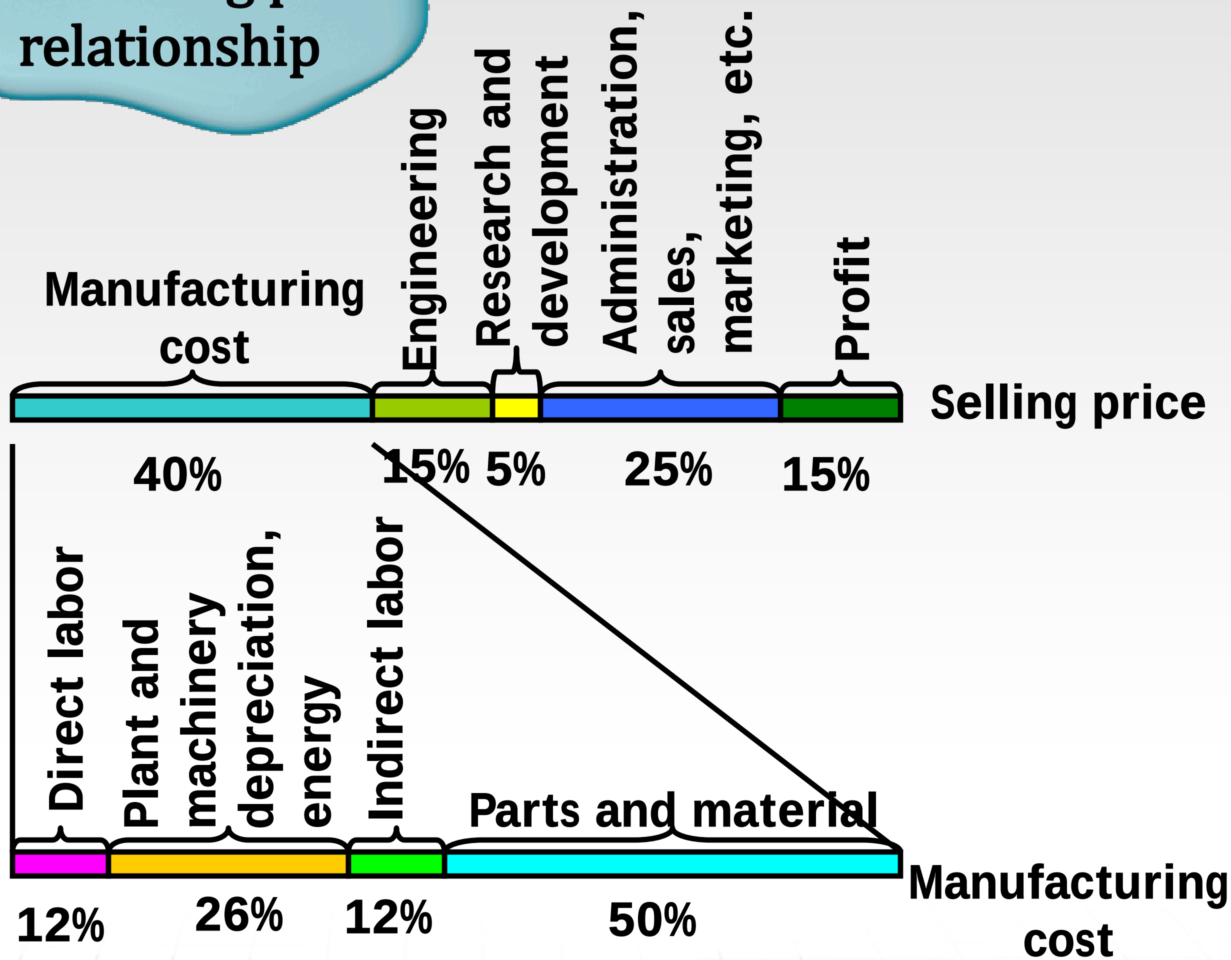
Work-in-process (WIP) is the quantity of parts or products currently located in the factory that are either being processed or are between processing operations.

$$\mathbf{WIP} = \frac{\mathbf{A \times U \times PC \times MLT}}{\mathbf{S \times H}}$$

where:

WIP – work-in-process in the facility (pc)
A – availability
U – utilization
PC – production capacity of the facility (pc/wk)
MLT – manufacturing lead time (hr)
S– number of shifts per week (shift/wk)
H – hours per shift (hr/shift)

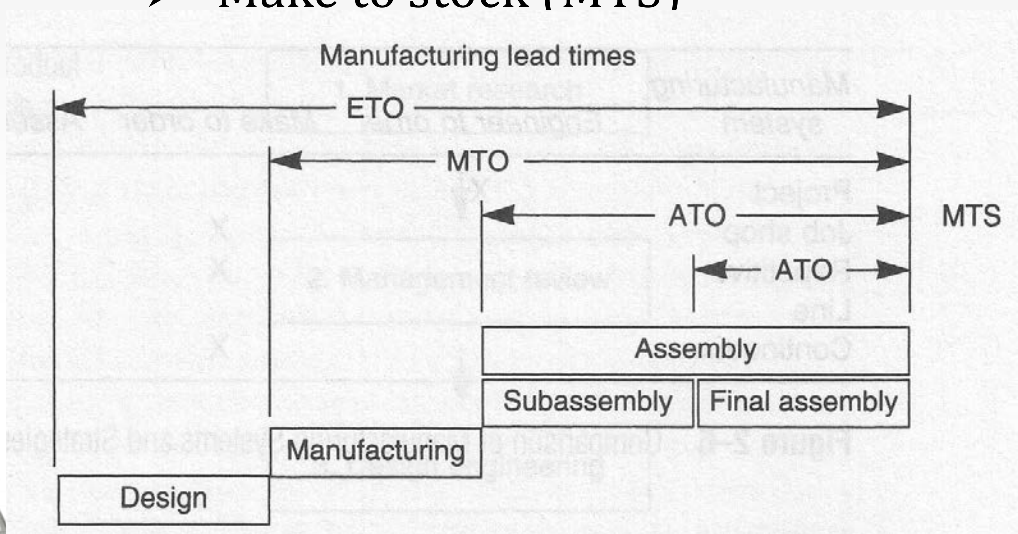
Manufacturing cost - selling price relationship

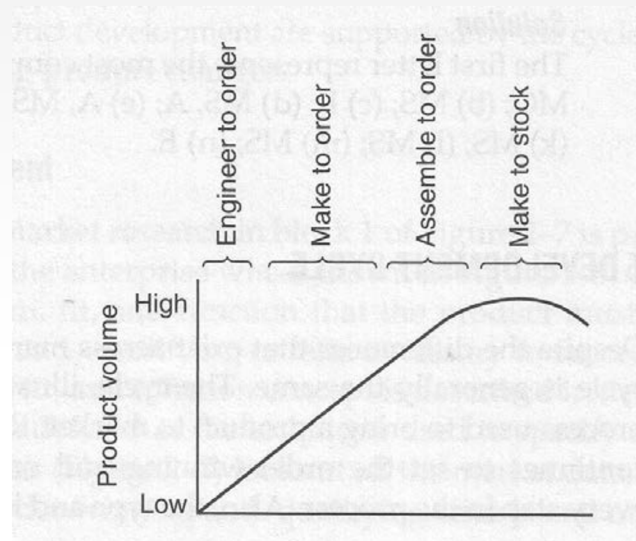


Black, J T., The Design of the Factory with a Future, McGraw-Hill, Inc., NY, 1991, provides some typical percentages for the different types of manufacturing and corporate .

Production strategy classification

- Engineer to order (ETO)
- Make to order (MTO)
- Assemble to order (ATO)
- Make to stock (MTS)





The production strategy used by manufacturers is based on several factors but the two most critical are: customer lead time and manufacturing lead time.

Engineer to Order

A manufacturer producing in the ETO category has either:

- a product that is in the first stage of the lifecycle curve or,
- a complex product with a unique design produced in single-digit quantities

The customer is willing to accept a long manufacturing lead time because the engineering design is part of the process.

Example: construction industry products (bridges, chemical plants, automotive production lines) and large products with special options that are stationary during production (commercial passenger aircraft, ships, high-voltage switchgear, steam turbines).

Make to Order

Manufacturers use this strategy when:

- the demand is unpredictable;
- the customer lead time permits the production process to start on receipt of an order.

The MTO technique assumes that all the engineering and design are complete and the production process is proven.

Example: new residential homes

Assemble to Order

The ATO technique is used when:

- the customer lead time is less than the manufacturing lead time,
- the option mix for the products can be forecast statistically,
- the subassemblies and parts for the final product are carried in a finished-components inventory.

Example: Automotive industry, IT industry, etc

Make to Stock

The MTO strategy is used for two reasons:

- the customer lead time is less than the manufacturing lead time,
- the product has a set configuration and few options so that the demand can be forecast accurately.

This strategy usually applies to mature products and commodity-type products such as toothpaste and food products.

Example: Toothpaste, breakfast cereals, etc.

Self-assessment test

1. Utilization of a machine is not equivalent with the availability of a machine. **T** **F**

2. Work-in-process are the components or products ready for delivery. **T** **F**

3. According to J.T. Black, the price of parts and materials is 50% from the selling price of a product. **T** **F**

4. Assemble to order is applied for the production of a car. **T** **F**

5. Make to order is applied when the demand can't be predicted. **T** **F**

Conclusions

- The classification of production processes helps us understand some of the similarities and differences that are present.

- Armed with this knowledge, a company can much more easily do the following:
  - ✓ Meet the external challenges
  - ✓ Identify the order-winning criteria
  - ✓ Implement the CIM principles
  - ✓ Adjust the management philosophy
  - ✓ Integrate the hardware and software systems
  - ✓ Compete in the world marketplace

- Material handling equipment
- Material transport systems
- Storage systems used in manufacturing
- Storage systems performance
- Storage strategies
- Automatic storage systems